

Report - Version 6: Feature Engineering

Version 6 extends **Building Agentic AutoML** with automatic feature generation and feature-space comparison.

The agent preserves task detection, feature detection, Data Inspection, the two preprocessing strategies, the three candidate models, and Smart Validation from Version 5. The new capability is to compare the original feature space with an engineered space containing **pairwise numerical interactions**.

Feature engineering is applied inside every fold after preprocessing. Models, metrics, validation, and hyperparameter configurations intentionally remain unchanged so that this version introduces only one new concept: **Feature Engineering**.

Goal of Version 6

- preserve the components introduced in Versions 1-5;
- add the **none** and **interactions** feature engineering strategies;
- generate pairwise products between numerical features while preserving original features;
- evaluate every preprocessing x feature engineering x model combination across 5 folds;
- store fold scores, mean, and standard deviation for each experiment;
- return best_preprocessing, best_feature_engineering, best_model, best_score, and best_std.

Architecture

```
Dataset + Target -> Task/Features -> Data Inspection -> Preprocessing  
Strategies -> Feature Engineering -> Model Candidates -> Smart Validation  
-> Evaluation -> Best Experiment -> State
```

Task Detection

Task detection remains unchanged: at most 20 unique target values means classification; otherwise the task is treated as regression.

```
def detect_task(df, target): n_unique = df[target].nunique() if n_unique <= 20:  
    return "classification" return "regression"
```

Feature Detection

Feature detection is also unchanged. On Adult Income, the agent detects 6 numerical features and 8 categorical features.

Data Inspection

The Data Inspector introduced in Version 2 is preserved unchanged. On Adult Income it reports 48,842 rows, 15 columns, two target classes, dtypes, missing values, and cardinality. The main missing values remain workclass = 2,799, occupation = 2,809, and native_country = 857.

Preprocessing

Version 6 preserves the **native** and **impute** strategies. Imputation parameters are learned exclusively from the training fold.

Feature Engineering

This is the new capability introduced in Version 6. **none** keeps the original feature space; **interactions** adds the product of every pair of numerical features. On Adult Income, 6 numerical features generate 15 additional interaction features. Original features are always preserved.

Model Selection

LightGBM, XGBoost, and CatBoost remain the three candidates introduced in Version 4. No hyperparameter search is performed.

Metric Selection

Metric selection remains unchanged: ROC AUC for classification and RMSE for regression.

Smart Validation

Validation from Version 5 remains unchanged: **StratifiedKFold** for classification and **KFold** for regression, both with 5 folds, shuffle=True, and random_state=42.

Training and Evaluation

Every preprocessing-feature engineering-model combination is evaluated across 5 folds. With 2 preprocessing strategies, 2 feature engineering strategies, and 3 models, the agent compares 12 experiments and performs 60 fits. Inside each fold: preprocessing -> feature engineering -> fresh model instance.

Results - Adult Income

Preprocessing	Feature Eng.	Model	Mean ROC AUC	Std
native	none	LightGBM	0.929556	0.002508
native	none	XGBoost	0.926851	0.002227
native	none	CatBoost	0.930631	0.002246
native	interactions	LightGBM	0.928578	0.002922
native	interactions	XGBoost	0.925359	0.002862
native	interactions	CatBoost	0.929786	0.002187
impute	none	LightGBM	0.929217	0.002713
impute	none	XGBoost	0.926435	0.002643
impute	none	CatBoost	0.930119	0.002428
impute	interactions	LightGBM	0.928319	0.002660
impute	interactions	XGBoost	0.925207	0.002604
impute	interactions	CatBoost	0.929030	0.002533

Best experiment: **CatBoost + native + none**, mean ROC AUC = **0.9306306263**, std = **0.0022463105**.

Agent

- load dataset and target;
- detect task and feature types;
- inspect the dataset;
- prepare X and y;
- select the metric and validation strategy;
- build candidate models;
- evaluate every preprocessing x feature engineering x model combination across 5 folds;
- store fold scores, mean, and standard deviation;
- select the best experiment using the mean score;
- return the final state.

```
{ "task": "classification", "metric": "roc_auc", "validation": "StratifiedKFold",  
  "n_splits": 5, "best_preprocessing": "native", "best_feature_engineering": "none",  
  "best_model": "CatBoost", "best_score": 0.930630626301291, "best_std":  
  0.002246310473216, "experiments": [...] }
```

Regression Test

The same agent is executed on `simple_regression.csv`. It automatically detects regression, selects KFold, uses RMSE, and evaluates all 12 preprocessing-feature engineering-model combinations.

Preprocessing	Feature Eng.	Model	Mean RMSE	Std
native	none	LightGBM	1.843523	0.107316
native	none	XGBoost	2.015096	0.065982
native	none	CatBoost	1.663274	0.103853
native	interactions	LightGBM	1.857629	0.117062
native	interactions	XGBoost	2.114515	0.127715
native	interactions	CatBoost	1.728558	0.113004
impute	none	LightGBM	1.856696	0.120398
impute	none	XGBoost	2.055328	0.130456
impute	none	CatBoost	1.628342	0.101940
impute	interactions	LightGBM	1.893113	0.114140
impute	interactions	XGBoost	2.051424	0.112415
impute	interactions	CatBoost	1.681678	0.112769

Best experiment: **CatBoost + impute + none**, mean RMSE = **1.6283421164**, std = **0.1019398476**.

What Version 6 Teaches

Version 6 introduces the idea that feature space itself should be treated as an AutoML choice and compared empirically. Explicit interactions do not improve the best result on either benchmark, but the agent can now generate, evaluate, and reject them when they are not useful.

The evolutionary pattern remains explicit: V1 builds the baseline, V2 observes the data, V3 experiments with preprocessing, V4 compares models, V5 makes validation more robust, and V6 compares feature spaces. The **One Version, One Concept** principle remains intact.

Limitations

- task detection still relies only on the number of unique target values;
- only two preprocessing strategies are available;
- feature engineering is limited to pairwise products between numerical features;
- the number of interaction features grows quadratically with numerical feature count;
- only three boosting model candidates are compared;
- cross-validation is fixed to 5 folds, with no group-aware or time-series-aware logic;
- no hyperparameter optimization;
- no planner or multi-agent architecture;
- the state does not yet store a persistent fitted pipeline.

Conclusion

Version 6 adds automatic comparison between original features and numerical interactions without changing preprocessing, models, metrics, validation, or hyperparameters. On both datasets the best strategy remains **none**, but the system gains a generalizable new capability and prepares the project for Version 7 Hyperparameter Optimization.